

# Diagnostic Evaluation Report

## 1. Framework Information

- \*\*Framework:\*\* X-RAG Diagnostic Framework (v1.0)
- \*\*Trace ID:\*\* `3fd5fd37-7dcd-40cf-869e-264b885734dd`
- \*\*Timestamp:\*\* 2026-09-05T14:27:16.266810Z
- \*\*Status:\*\* COMPLETED

## 2. Executive Summary

- \*\*Question:\*\* According to the BNSS, what is the specific process and timeline for filing a trademark infringement suit?
- \*\*Answer:\*\* I do not have enough information to answer this.
- \*\*Health Score:\*\* 0.08
- \*\*Primary Issue:\*\* RETRIEVAL\_MISS
- \*\*Claims Supported:\*\* 0 / 1
- \*\*Summary:\*\* Analysis completed.

## 3. Pipeline Overview

| Stage     | Status  | Confidence | Observation                                                                                     |
|-----------|---------|------------|-------------------------------------------------------------------------------------------------|
| ---       | ---     | ---        | ---                                                                                             |
| CORPUS    | UNKNOWN | 0.5        | Corpus relevance is inconclusive.                                                               |
| RETRIEVER | FAIL    | 0.85       | No retrieved chunks provide support, and retrieval scores strongly indicate a lack of useful ev |
| GENERATOR | UNKNOWN | 0.5        | 1 generated claims are unsupported, but retrieval quality is ambiguous.                         |
| GROUNDING | PASS    | 0.9        | No claims are contradicted by retrieved evidence.                                               |

\*Efficiency: 0/6 chunks used (0%)\*

## 4. Evaluation Metrics

- \*\*Retrieved Chunks:\*\* 6
- \*\*Verified Claims:\*\* 1 (Supported: 0, Partial: 0, Unsupported: 1, Contradicted: 0)
- \*\*Grounding Score:\*\* 0.00
- \*\*Evidence Coverage:\*\* 0.17
- \*\*Average Entailment:\*\* 0.04
- \*\*Latencies:\*\* Retrieval (2015.2ms) | Generation (6779.1ms) | Verification (4093.9ms)

## 5. RAGAS Metrics

- \*\*Faithfulness:\*\* 0.0000
- \*\*Answer Relevancy:\*\* 0.3440
- \*\*Context Precision:\*\* 0.0000
- \*\*Context Relevancy:\*\* 0.0000
- \*\*Context Recall:\*\* N/A
- \*\*Answer Similarity:\*\* N/A
- \*\*Answer Correctness:\*\* N/A

## Answer Correctness (Claim Recall)

Not computed for this trace.

## 6. Evidence Analysis

- \*\*[7d568793-fd79-485a-ba6a-3cb984facaoa9]\*\* (UNSUPPORTED): I do not have enough information to answer this.
- \*Evidence (Chunk 78931bc9-967c-48ea-bf5d-331f90834436):\* 1]

THE GAZETTE OF INDIA EXTRAORDINARY  
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## 7. Root Cause Analysis

- \*\*Primary Cause:\*\* RETRIEVAL\_MISS
- \*\*Confidence:\*\* 0.88
- \*\*Reasoning:\*\*

- Starting root cause analysis for trace 3fd5fd37-7dcd-40cf-869e-264b885734dd.
- Skipping CORPUS: status is UNKNOWN (insufficient evidence).
- Failure at RETRIEVER: RETRIEVAL\_MISS (confidence=0.85). Observation: No retrieved chunks provide support, and retrieval
- Skipping CHUNKING: no observable state in matrix.
- Skipping GENERATOR: status is UNKNOWN (insufficient evidence).
- GROUNDING passed (confidence=0.90).
- Primary root cause: RETRIEVAL\_MISS at RETRIEVER (earliest upstream failure, confidence=0.85).
- Analysis complete. Primary cause: RETRIEVAL\_MISS.

## ## 8. Corrective Action Plan

- **\*\*[IMMEDIATE] Increase Retrieval Top-K\*\***
  - Description: Increase the number of retrieved documents before providing context to the LLM.
  - Expected Improvement: Higher recall; missing context is caught within the expanded window. (Metric: Verification support)
- **\*\*[SHORT\_TERM] Implement Hybrid Search (Dense + Sparse)\*\***
  - Description: Add BM25 lexical search and combine with semantic vector search using Reciprocal Rank Fusion.
  - Expected Improvement: Near-perfect recall for exact-match terminology. (Metric: Retrieval hit rate for exact entity quer)
- **\*\*[EXPERIMENTAL] Deploy Cross-Encoder Reranker\*\***
  - Description: Pass the Top-N results through a Cross-Encoder to re-score query-document relevance.
  - Expected Improvement: Drastically improved precision at Top-1 and Top-3. (Metric: Precision@3 increases by at least 20%)
- **\*\*[INFORMATIONAL] Review Retrieval Top-K / Reranker Cutoff\*\***
  - Description: Retrieval efficiency is low (0%); consider reducing top-k or tightening the reranker score cutoff.
  - Expected Improvement: Reduced prompt size and lower latency/cost with no loss of verified support. (Metric: Chunk utiliz

## ## 9. Evidence Traceability

- **\*\*Primary Cause:\*\*** RETRIEVAL\_MISS
- **\*\*Derived From:\*\*** RootCauseAnalysis
- **\*\*Pipeline Stage:\*\*** RETRIEVAL\_MISS
- **\*\*Supporting Claims:\*\*** 7d568793-fd79-485a-ba6a-3cb984faca9
- **\*\*Supporting Chunks:\*\*** 78931bc9-967c-48ea-bf5d-331f90834436

## ## 10. Overall Assessment

- **\*\*Major Strength:\*\*** GROUNDING is healthy (confidence=90%): No claims are contradicted by retrieved evidence.
- **\*\*Major Weakness:\*\*** RETRIEVAL\_MISS
- **\*\*Next Priority:\*\*** Increase Retrieval Top-K
- **\*\*Recommendation:\*\*** Pipeline requires architectural adjustments across multiple stages.

## ## 11. Framework Footer

- **\*\*Generated By:\*\*** X-RAG Diagnostic Framework
- **\*\*Artifact Version:\*\*** 1.0

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## ## APPENDIX

### Artifacts Generated

#### ✓ RAGTrace

artifacts/rag\_traces/3fd5fd37-7dcd-40cf-869e-264b885734dd.json

#### ✓ ClaimSet

artifacts/claim\_sets/3fd5fd37-7dcd-40cf-869e-264b885734dd.json

#### ✓ VerificationResults

artifacts/verification/3fd5fd37-7dcd-40cf-869e-264b885734dd.json

#### ✓ PipelineStateMatrix

artifacts/pipeline\_state\_matrix/3fd5fd37-7dcd-40cf-869e-264b885734dd.json

#### ✓ RootCauseAnalysis

artifacts/root\_cause\_analysis/3fd5fd37-7dcd-40cf-869e-264b885734dd.json

#### ✓ CorrectiveActionPlan

artifacts/corrective\_action\_plan/3fd5fd37-7dcd-40cf-869e-264b885734dd.json

✓ DiagnosticEvaluationReport

artifacts/reports/3fd5fd37-7dcd-40cf-869e-264b885734dd.json